

IISER Kolkata
Department of Mathematics & Statistics



On Flatness and Controllability of the Heat Equation in One Dimension

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August 5, 2021

Outline

1. Introduction
2. Flatness Method for ODEs
3. Flatness Approach for Heat Equation

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Flatness Approach

A Brief Overview

- Introduced in the nineties by M. Fliess, J. Lévine, P. Martin and P. Rouchon.
  - Motion Planning of systems governed by ODEs (See []).
- In the twenties, it was then adapted to some PDEs.
  - Heat Equation (See [MRR14, LMR00]), Beam Equation (See []), Korteweg de Vries equation (See [MRRR19]), Schrodinger equation (See [MRR19b]). See also [MRR18a]
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An Abstract Differential System

Consider the control system

$$\frac{du}{dt} = f(u, q), \tag{1}$$

where

- $u \in \mathbb{R}^n$ is the state
- $q \in \mathbb{R}^m$ is the input (the control).

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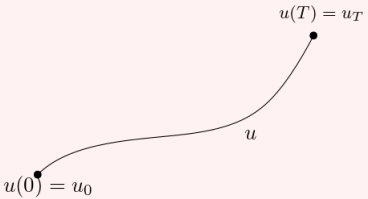


Figure: Exact Controllability

ODE System

Consider a smooth control system

$$u'(t) = f(u, q), \quad u \in \mathbb{R}^n, \quad q \in \mathbb{R}^m \tag{2}$$

with an output $y \in \mathbb{R}^s$ such that

$$y = h\left(u, q, q', \cdots, q^{(r)}\right). \tag{3}$$

- We say y is a flat output if

$$u = g\left(y, y', \cdots, y^{(p)}\right), \tag{4}$$

$$q = h\left(y, y', \cdots, y^{(q)}\right), \tag{5}$$

where $p, q \in \mathbb{N} \cup \{0\}$ and g, h are some smooth functions.

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Motion Planning for ODE Systems

Consider the control problem

$$\begin{aligned} \frac{du}{dt} &= f(u, q), \\ u(0) = u_0, &\quad u(T) = u_T. \end{aligned} \tag{6}$$

Problem. Find a flat output y and a smooth function g such that

$$u(0) = g\left(y, y', \cdots, y^{(p)}\right)(0) = u_0, \tag{7}$$

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An Illustrative Example

Consider the linearized pendulum

$$\begin{aligned} u_1'(t) &= u_2(t), \\ u_2'(t) &= q(t), \end{aligned} \tag{9}$$

where $u = (u_1, u_2) \in \mathbb{R}^2$ is the state and $q \in \mathbb{R}$ is the control.

- Choose an output $y(t) = u_1(t), \quad t \geq 0$
- $u_1 = y$, $u_2 = y'$ and $q = y'' \implies$ The output y is flat

Choose the initial and final state $u_0 = (0, 0)$ and $u_T = (1, 0)$. We have to find $y \in C^\infty([0, T], \mathbb{R})$ such that

$$y(0) = 0, \quad y'(0) = 0, \quad y(T) = 1, \quad y'(T) = 0. \tag{10}$$

One suitable choice is

$$y(t) = \frac{t^2(2T - t)^2}{T^4}, \quad t \in [0, T]. \tag{11}$$

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The Heat Equation in One Dimension

The heat equation in the interval $(0, 1)$ is

$$u_t(t, x) - u_{xx}(t, x) = 0 \tag{12}$$

with $t \in (0, T)$. We consider the following boundary and initial conditions:

$$\begin{aligned} &u_x(t, 0) = 0, \quad u_x(t, 1) = q(t), \ t \in (0, T), \\ &u(0, x) = u_0(x), \quad x \in (0, 1). \end{aligned} \tag{13}$$

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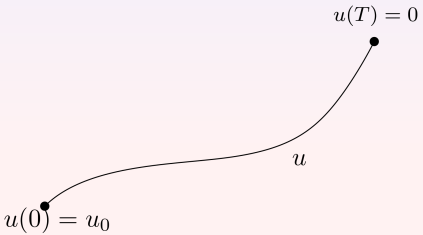


Figure: Null Controllability

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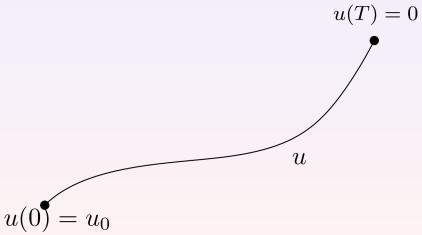
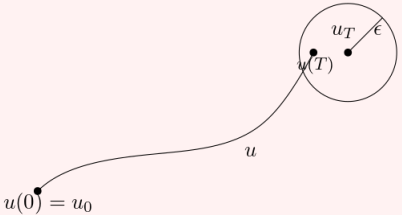
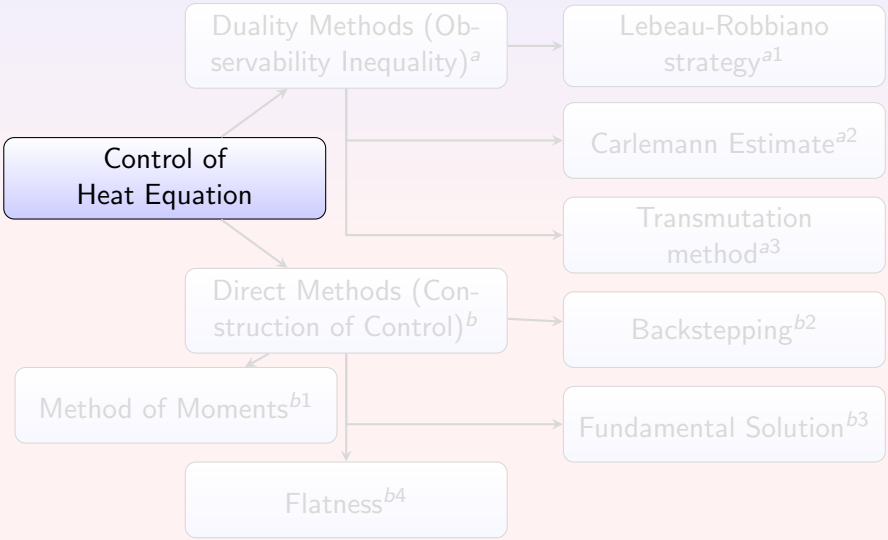


Figure: Null Controllability

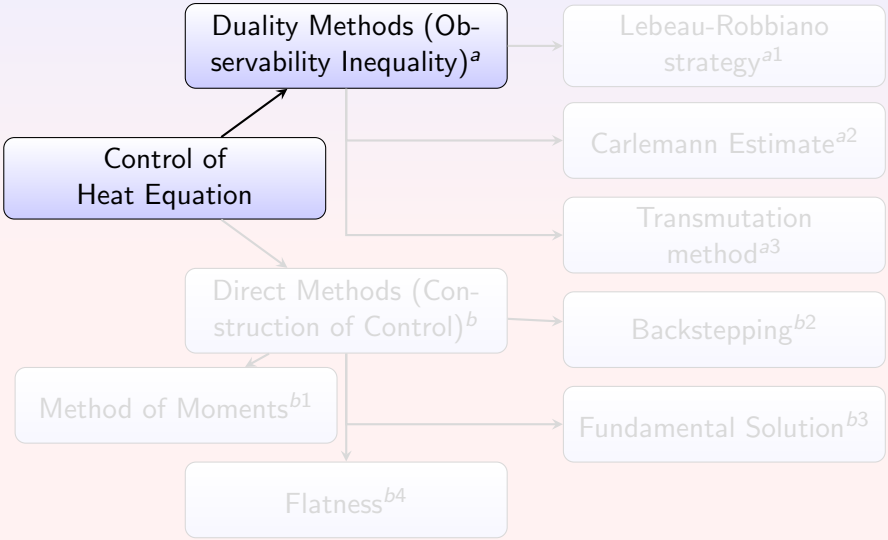


Control of Heat Equation (1D)

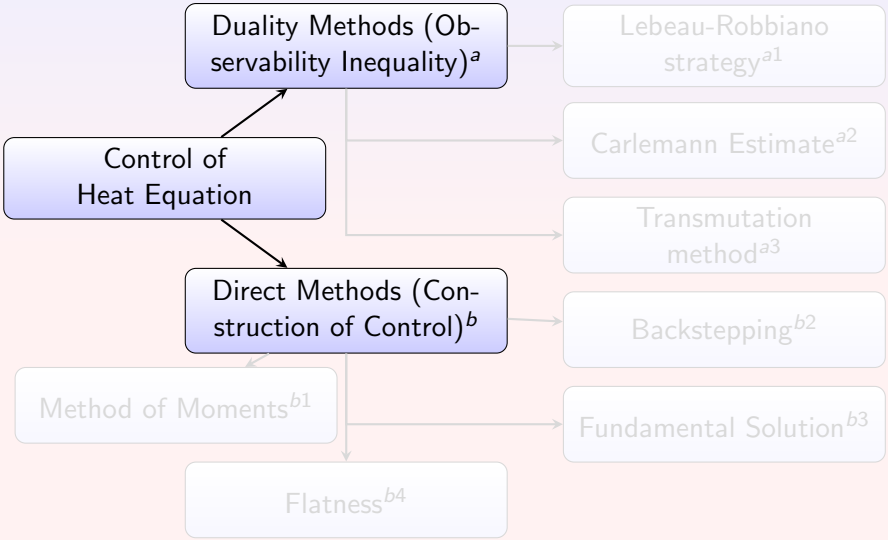
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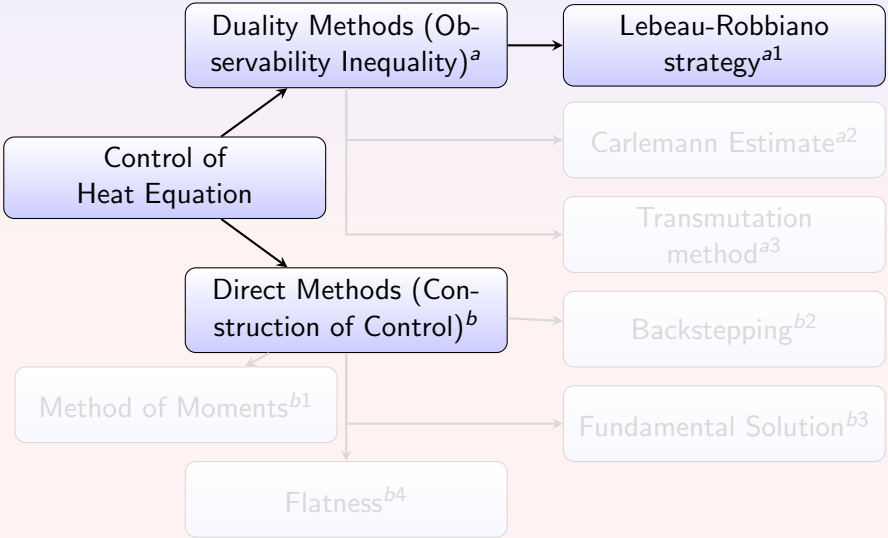
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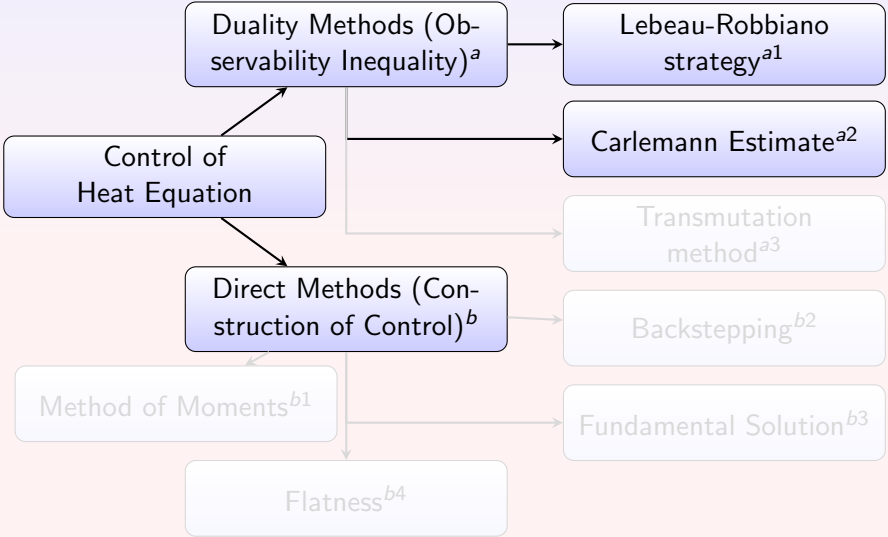


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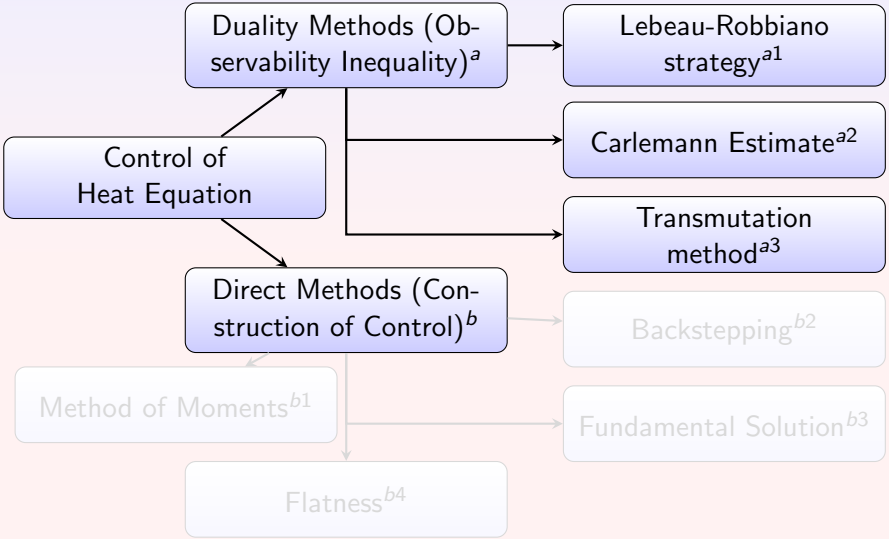
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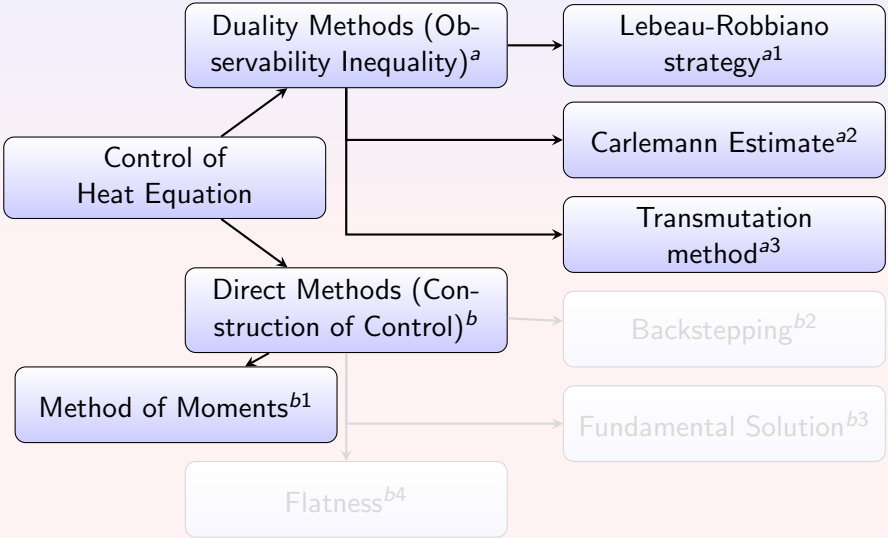
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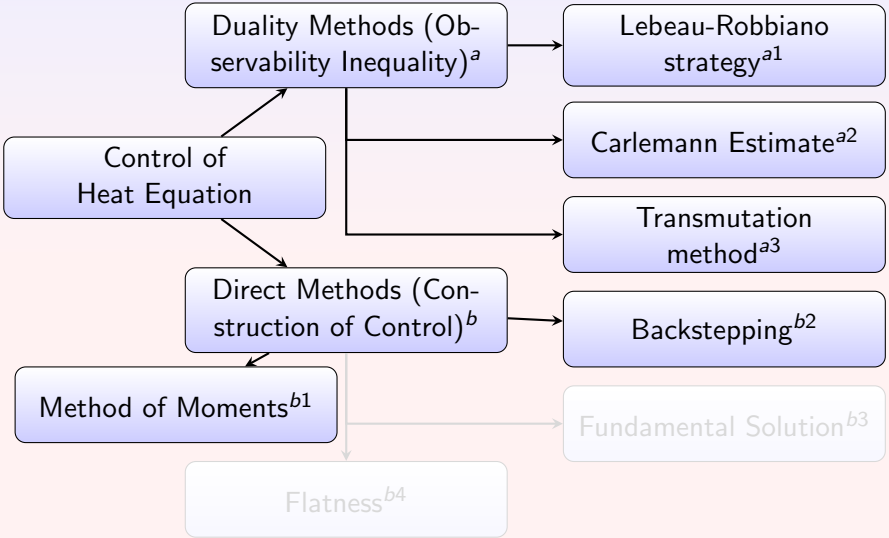
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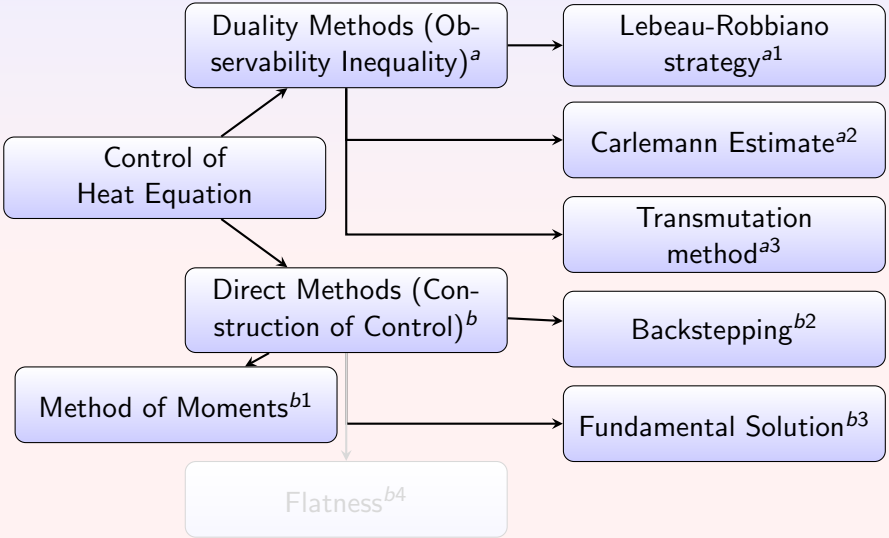
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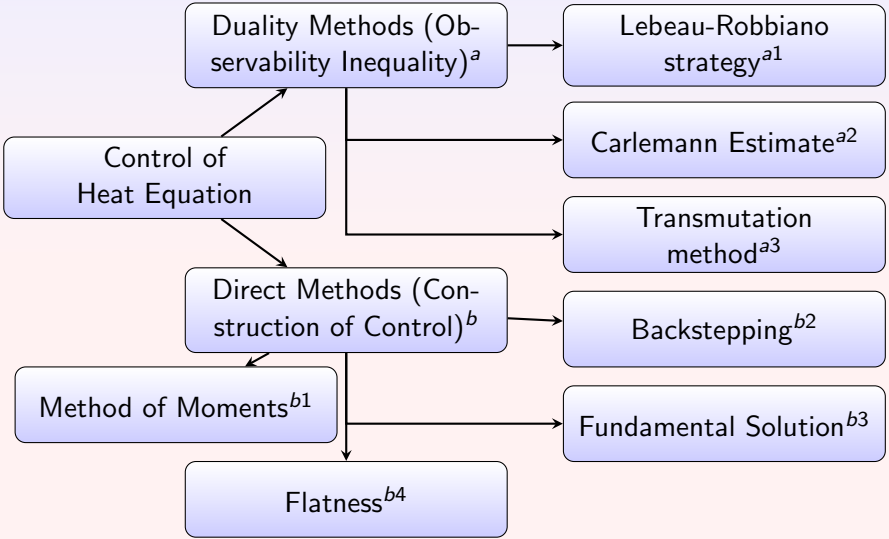
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Control of Heat Equation (1D)



$b3$. Littman, Walter, Boundary control theory for hyperbolic and parabolic partial differential equations with constant coefficients, Ann. Scuola

Control of Heat Equation (1D)



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Trajectory Planning

- We find solutions to equation (12) of the form

$$u(t,x) = \sum_{k\geq 0} u_k(t)\frac{x^k}{k!}, \quad t \in (0,T), \quad x \in (0,\pi). \tag{14}$$

- We get

$$u_{k+2}(t) = u_k(t). \tag{15}$$

for all $k \geq 0$.

- Define a function $y : [0,\infty) \to \mathbb{R}$ (called the flat output) such that

$$y(t) = u(t,0), \quad t \geq 0 \tag{16}$$

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- We find solutions to equation (12) of the form

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$$u_{k+2}(t) = u_k'(t), \tag{15}$$

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Trajectory Planning contd.

- Boundary condition implies the explicit expressions of u_k for $k \geq 0$

$$u_{2k+1}(t) = 0, \quad \& \quad u_{2k}(t) = y^{(k)}(t), \tag{17}$$

for all $k \geq 0$ and $t \in (0, T)$.

- Thus the formal solution is

$$u(t, x) = \sum_{k=0}^{\infty} y^{(k)}(t) \frac{x^{2k}}{(2k)!} \tag{18}$$

for $x \in (0, 1)$ and $t \in (0, T)$.

- The formal control is

$$q(t) = \sum_{k=1}^{\infty} \frac{y^{(k)}(t)}{(2k-1)!} \tag{19}$$

for $t \in (0, T)$.

Trajectory Planning contd.

- Boundary condition implies the explicit expressions of u_k for $k \geq 0$

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for all $k \geq 0$ and $t \in (0, T)$.

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Gevrey Functions

Definition (Gevrey Function (One Variable))

We say a C^∞ function $t \in [0, T] \mapsto y(t)$ is Gevrey of order α if there exist $M, R > 0$ such that

$$\sup_{t\in[0,T]} \left| y^{(m)}(t) \right| \leq M \frac{(m!)^\alpha}{R^m}, \quad \text{for all } m \in \mathbb{N}. \tag{20}$$

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- A Gevrey function of order α is also of order β for any $\beta \geq \alpha$.
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Examples of Gevrey Functions

Example

Let $\gamma > 0$. The functions

$$\phi_\gamma(t) = \begin{cases} 0, & \text{when} \quad t \notin (0,1), \\ e^{\frac{-1}{((1-t)t)^\gamma}}, & \text{when} \quad t \in (0,1) \end{cases} \tag{21}$$

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Convergence of the solution

Theorem

Let y be Gevrey of order $1 \leq \alpha < 2$. The formal solution (18) is Gevrey of order α in t and order $\frac{\alpha}{2}$ in x. In particular, the formal control q is Gevrey of order α.
Furthermore, if $\alpha = 2$ the same result holds provided $R > 4$.

Proof.

Note that for all $(t, x) \in [0, T] \times [0, 1]$,

$$\left| \frac{y^{(k+m)}(t)}{m!^{\alpha} n!} \frac{x^{2k-n}}{(2k-n)!} \right| \leq M \frac{(\pi i)^{\frac{\alpha}{4}}}{R^i (2i-n)!^{1-\frac{\alpha}{2}}} n!^{\frac{\alpha}{2}} \frac{m!^{\alpha}}{R^m},$$

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Null Controllability

Theorem

Let $T > 0$, $0 < \tau < T$ and $1 < \alpha < 2$. For every initial state $u_0 \in L^2(0,1)$ there is a function $y : [\tau, T] \to \mathbb{R}$ Gevrey of order α on $[\tau, T]$ such that the control

$$q(t) = \begin{cases} 0, & 0 \leq t \leq \tau, \\ \sum_{k \geq 1} \frac{y^{(k)}(t)}{(2k-1)!}, & \tau < t \leq T \end{cases} \tag{23}$$

steers the control system (12) from the initial state $u(0) = u_0$ to the final state $u(T) = 0$ at time T.

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Step 1. Regularization of Initial State

The Control $u(t) = 0$ on $[0, \tau]$

- We decompose the initial state $w_0 \in L^2(0,1)$ as

$$w_0(x) = \sum_{n \geq 0} c_n \sqrt{2} \cos(n\pi x),$$

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- The solution to the heat equation (12) is

$$w(t,x) = \sum_{n \geq 0} c_n e^{-n^2 \pi^2 t} \sqrt{2} \cos(n\pi x) \tag{24}$$

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Let $\tau \in (0, T)$. The mapping $x \mapsto u(\tau, x)$ is real analytic. Moreover, u is Gevrey of order α in t and $\frac{\alpha}{2}$ in x on $[\epsilon, \tau] \times [0, 1]$ for all $\epsilon \in (0, \tau)$.

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The proof follows from the following estimate. For all $k \geq 0$

$$|y_k| \leq C \left\| u_0 \right\|_{L^2(0,1)} \left(1 + \tau^{-\frac{1}{2}} \right) \frac{k!}{\tau^k}.$$

Step 2. Construction of the Control

Theorem

Let $0 < \tau < T$ and $1 < \alpha < 2$. The function

$$y(t) = \phi_\alpha\left(\frac{t-\tau}{T-\tau}\right)\sqrt{2}\sum_{n\geq 0}\alpha_n e^{-n^2\pi^2 t} \tag{25}$$

is Gevrey of order α on $[\tau, T]$.Moreover, for all $k \geq 0$

- $y^{(k)}(\tau) = y_k$
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- $\left|y^{(k)}(t)\right| \leq C \sup_{n\geq 0}|c_n| \frac{k!^\alpha}{R^\alpha}, \quad \forall t \in [\tau, T]$, *for some* $C, R > 0$.

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Existence of y satisfying $y^{(k)}(T) = 0$ for all $k \geq 0$ is impossible to do with an analytic function, but possible with a function Gevrey of order $\alpha > 1$.

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- The function

$$\tilde{y}(t) = \sqrt{2}\sum_{n\geq 0} \alpha_n e^{-n^2\pi^2 t}$$

is analytic on $(0, \infty)$ and hence Gevrey of order α on $[\tau, T]$.

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Proving Null Controllability

- The solution u on $[\tau, T]$ satisfies

$$u(\tau) = u_\tau, \quad \text{and} \quad u(T) = 0. \tag{28}$$

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Approximate Controllability

- Take $u_0 \in L^2(0,1)$ (initial state) and $u_T \in L^2(0,1)$ (final state).
- For all $\epsilon > 0$, there exist polynomials

$$p_0(x) = \sum_{k=0}^{N} p_k^0 \frac{x^{2k}}{(2k)!} \quad \text{and} \quad p_T(x) = \sum_{k=0}^{N} p_k^T \frac{x^{2k}}{(2k)!} \tag{30}$$

such that

$$\|u_0 - p_0\|_{L^2(0,1)} \leq \epsilon, \quad \text{and} \quad \|u_T - p_T\|_{L^2(0,1)} \leq \epsilon. \tag{31}$$

- The function

$$y_M(t) = \sum_{k=0}^{M} \left[p_k^0 \frac{t^{2k}}{k!} \left(1 - \phi_\epsilon\left(\frac{t}{T}\right)\right) + p_k^T \frac{(t-T)^{2k}}{k!} \phi_\epsilon\left(\frac{t}{T}\right) \right] \tag{32}$$

is Gevrey of order ≤ 2 when $\epsilon \geq 1$.

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$$p_0(x) = \sum_{k=0}^{N} p_k^0 \frac{x^{2k}}{(2k)!}, \quad \text{and } p_T(x) = \sum_{k=0}^{N} p_k^T \frac{x^{2k}}{(2k)!} \tag{30}$$

such that

$$\|u_0 - p_0\|_{L^2(0,1)} \leq \epsilon, \quad \text{and } \|u_T - p_T\|_{L^2(0,T)} \leq \epsilon. \tag{31}$$

- The function

$$y_N(t) = \sum_{k=0}^{N} \left[p_k^0 \frac{t^k}{k!} \left(1 - \Phi_\gamma\left(\frac{t}{T}\right)\right) + p_k^T \frac{(t-T)^k}{k!} \Phi_\gamma\left(\frac{t}{T}\right) \right] \tag{32}$$

is Gevrey of order ≤ 2 when $\gamma \geq 1$.

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Contd.

- y_N satisfies

$$\begin{aligned} {y_N}^{(k)}(0) &= p_k^0, \quad {y_N}^{(k)}(T) = p_k^T, \quad k = 0, 1, \cdots, n \\ {y_N}^{(k)}(0) &= 0, \quad {y_N}^{(k)}(T) = 0, \quad k > n. \end{aligned} \tag{33}$$

- Define the open loop control

$$q_N(t) = \sum_{k \geq 0} \frac{{y_N}^{(k)}(t)}{(2k)!}, \quad t \in [0, T]. \tag{34}$$

- Consider an approximate control $\bar{q_N} \in C^2(0, T)$ such that

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Approximate Controllability of the Heat Equation

Theorem

Let $\epsilon > 0$, $T > 0$. For any $u_0, u_T \in L^2(0,1)$, the control $\bar{q}_N$ approximately steers the system (12) from the initial state u_0 to the final state u_T in time T, that is,

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where $C(T)$ is a constant depends only on T.

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Proof

- Define $\bar{u}(t,x) = u(t,x) - q(t)$ for $(t,x) \in (0,T) \times (0,1)$.
- The system (12) is then transforms to interior control system

$$\begin{aligned} &\bar{u}_t(t,x) - \bar{u}_{xx}(t,x) = -q'(t) \\ &\bar{u}_x(t,0) = 0, \quad \bar{u}(t,1) = 0, \quad t > 0 \\ &\bar{u}(0,x) = u_0(x) - q(0) = \bar{u}_0(x), \quad x \in (0,1). \end{aligned} \tag{37}$$

- Let A be the operator defined by

$$\begin{aligned} D(A) &= \{f \in L^2(0,1) : f_x \in L^2(0,1),\ f_x(0) = 0,\ f(1) = 0\} \\ A(f) &= -f_{xx} \end{aligned}$$

Proof

- Define $\tilde{u}(t,x) = u(t,x) - q(t)$ for $(t,x) \in (0,T) \times (0,1)$.
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Proof contd.

- Hille-Yosida theorem implies existence of a unique solution in

$$C^1([0,T];L^2(0,1)) \cap C([0,T];L^2(0,1)) \cap C([0,T];D(A))$$

- We have

$$w(t,x) = S(t)p_0 - \int_0^t S(t-s)q(s)ds, \tag{38}$$

where $\{S(t)\}_{t\geq 0}$ denotes the semigroup of contraction associated with A and $\|S(t)\| \leq e^{-t}$.

- The function

$$v(t,x) = \sum_{k=0}^{2n} y_N^{(k)}(t)\frac{x^{2n}}{(2n)!} \tag{39}$$

is the unique solution of (37) with control $q = q_N$ and $v(0,x) = p_0 - q_N(0)$, $v(T,x) = p_T - q_N(T)$.

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- Hille-Yosida theorem implies existence of a unique solution in

$$C^1((0,T];L^2(0,1)\cap C([0,T];L^2(0,1)\cap C((0,T];\mathcal{D}(\mathcal{A}))$$

- We have

$$\tilde{u}(t,x)=S(t)\tilde{u}_0-\int_0^t S(t-s)q'(s)ds, \tag{38}$$

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Proof contd.

- We have

$$p_T - q_N(T) = S(T)(p_0 - q_N(0)) - \int_0^T S(T-s) q_N'(s) ds. \tag{40}$$

- This implies

$$u(T) - \bar{q}_N(T) = p_T - q_N(T) + S(T)(p_0 - p_0 + \Delta q_N(s)) - \int_0^T S(T-s) \Delta q_N'(s) ds \tag{41}$$

with $\Delta q_N = \bar{q}_N - q_N$.

Proof contd.

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Proof contd.

- Therefore

$$\begin{aligned}
&\|u(T,\cdot) - u_T(\cdot)\|_{L^2(0,1)} \\
&\leq \|p_T - u_T(\cdot)\|_{L^2(0,1)} + \|u(T,\cdot) - p_T(\cdot)\|_{L^2(0,1)} \\
&\leq \epsilon + \epsilon + 2\epsilon e^{-\omega T} + \frac{1 - e^{-\omega T}}{\omega}\epsilon \\
&\leq \left(4 + \frac{1}{\omega}\right)\epsilon
\end{aligned}$$

This completes the proof.

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- Therefore

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&\quad + \|S(T)\| \, \|u_0 - p_0 + \Delta q_N(0)\|_{L^2(0,1)} \\
&\quad + \sup_{t\in[0,T]} \left|\Delta {q_N}'(t)\right| \int_0^T \|S(T-s)\| \, ds \\
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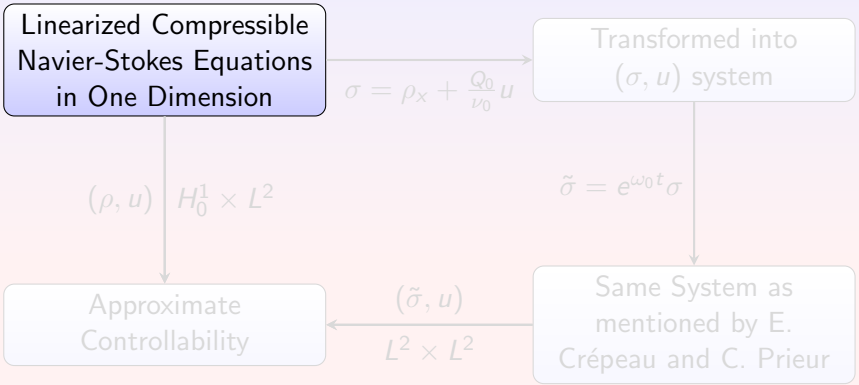
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Linearized Compressible Navier-Stokes Equations

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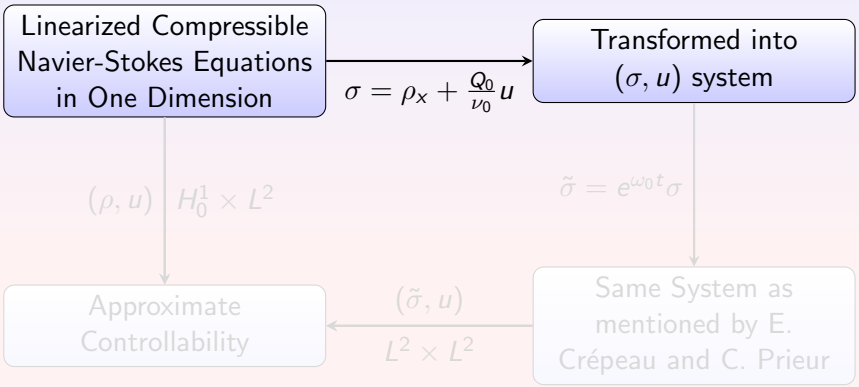
$$\rho_t + Q_0 u_x = 0,$$

$$u_t - \nu_0 u_{xx} + b\rho_x = 0,$$

$$u_x(t,0) = 0, u_x(t,1) = q(t),$$

$$\rho(0,x) = 0 = u(0,x).$$

Linearized Compressible Navier-Stokes Equations



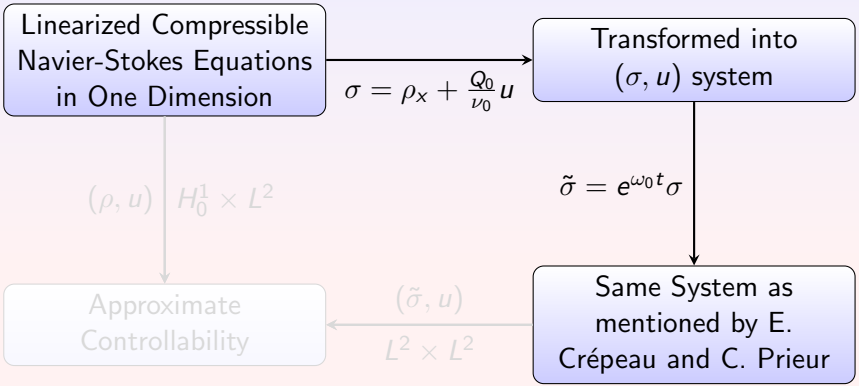
$$\sigma_t + \frac{bQ_0}{\nu_0}\sigma = \frac{bQ_0^2}{\nu_0^2} u,$$

$$u_t - \nu_0 u_{xx} + b\sigma = \frac{bQ_0}{\nu_0} u,$$

$$u_x(t,0) = 0, \quad u(t,1) = q(t),$$

$$\sigma(0,x) = 0, \quad u(0,x) = 0$$

Linearized Compressible Navier-Stokes Equations

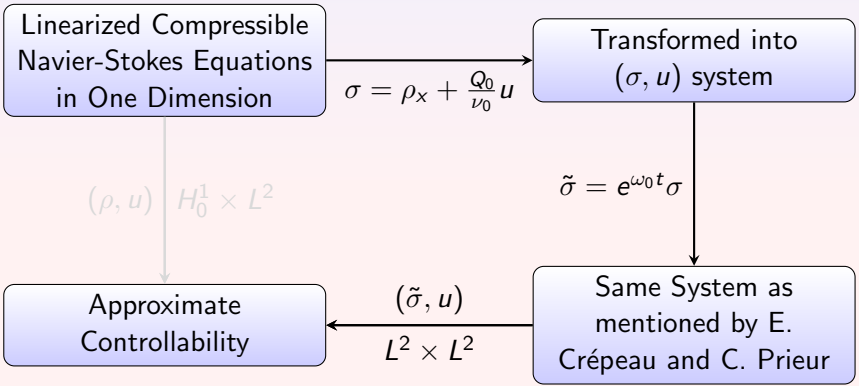


$$\tilde{\sigma}_t = \frac{bQ_0^2}{\nu_0^2}\tilde{u},$$

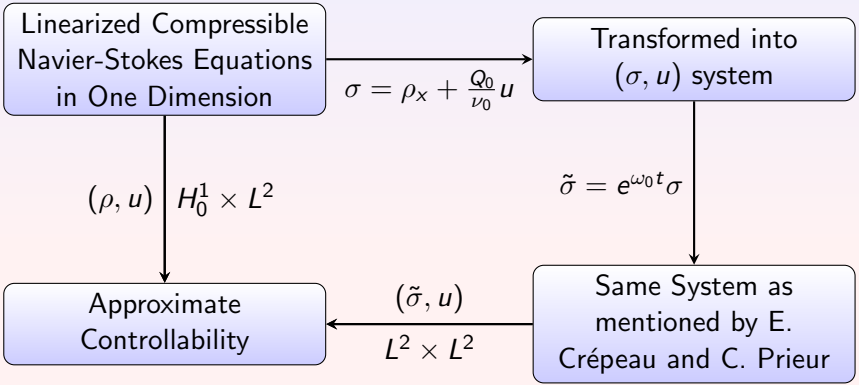
$$\tilde{u}_t - \nu_0 u_{xx} - b\tilde{\sigma} = 2\frac{bQ_0}{\nu_0}\tilde{u},$$

$$\tilde{u}_x(t,0) = 0, \quad \tilde{u}(t,1) = \tilde{q}(t),$$

Linearized Compressible Navier-Stokes Equations



Linearized Compressible Navier-Stokes Equations



$$\|\rho(T,\cdot) - \rho_T\|_{L^2(0,1)} \leq \|\rho_x(T,\cdot) - (\rho_T)_x\|_{L^2(0,1)} \quad \text{(Poincaré Inequality)}$$
$$\leq \|\sigma(T,\cdot) - \sigma_T\|_{L^2(0,1)} < \epsilon.$$

Usefulness of Flatness Method

- Characterization of Reachable Sets for PDEs (Heat Equation, KdV Equation etc.).
- Various Numerical aspects (For Linear and Nonlinear PDEs).
- Existence of kernels occur in Backstepping Method.
- Existence of kernels occur in Transmutation Method.
- Construction of Tychonov solution with coefficients in some Gevrey class.

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